Lumbini City College

Bachelor Level / Fifth Semester / Science — **Full Marks:** 60
Computer Science and Information Technology (CSC316) — **Pass Marks:** 24
Cryptography — **Time:** 3 Hours

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Section A

Attempt any TWO questions. **[10 x2 = 20]**

1) Define the CIA Triad and explain its components. List the security services. Differentiate between block ciphers and stream ciphers with examples.

2) Define discrete logarithm. Explain key generation, encryption, and decryption in RSA.
Given $p = 61$, $q = 53$, encrypt and decrypt the message $M = 42$.

3) Explain the Feistel cipher structure. Describe the single round operation in DES and the F-function.

Section B

Attempt any EIGHT questions. **[8 x 5 = 40]**

4) Encrypt the message **"DOOMSTAY"** using Playfair cipher with the keyword "SECURITY".

5) Show encryption of the message **"MONA"** using Hill cipher with key

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

6) Explain digital signatures. Differentiate between direct and arbitrated digital signatures.

7) Define firewall. Explain different types of firewalls. Also list the elements of X.509.

8) Explain the Diffie-Hellman key exchange process.

9) What is a Denial of Service (DoS) attack? Explain with examples. Differentiate between worms, viruses, and Trojan horses.

10) Explain IPSec and its components. List the services provided by PGP.

11) Distinguish between substitution ciphers and transposition ciphers. Encrypt the message **"|I AM BECOME DEATH, THE DESTROYER OF WORLDS"** using Rail Fence cipher (rail size = 3).

12) State Fermat's Little Theorem with example. Explain Miller-Rabin primality test.

13) Explain the concept of authentication system. Describe one-way and mutual authentication.